

GeoAttend

IoT-Enabled Smart Attendance Management System

With Real-Time Geofencing, Cryptographic Device Fingerprinting, and Live Telemetry Analytics

A comprehensive technical design, implementation, and security reference manual detailing system features built from starting to final production release.

DEVELOPMENT MANUAL & TECHNICAL REFERENCE

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Project Overview & Architecture

1. Abstract & Motivation

Traditional attendance tracking (manual roll calls, signature lists, static QR codes, or basic card readers) fails to guarantee actual presence. Students often scan QR codes from screenshots sent by friends, share cards, or register check-in and immediately bunk class. GeoAttend addresses these issues by locking attendance to physical presence, mobile device fingerprint, and local biometric credentials.

Through integrated GPS geofencing, rotating time-varying tokens, stable browser fingerprinting, and cryptographic WebAuthn biometrics, the system establishes a zero-proxy operational baseline. Furthermore, continuous background telemetry ensures students remain within designated boundaries during college hours, alerting coordinators immediately upon exit.

2. Role-Based Permissions & Modules

HOD Console (Administrative Control)

Complete control over geofence bounds, operation hours, digital out-passes, holiday lock controls, and live analytics dashboards.

Coordinator Panel (Area Management)

Manages daily attendance rosters, handles keypad phone manual OTP codes, and monitors active GPS dropout alerts.

Teacher Dashboard (Classroom Monitor)

Launches classroom sessions, sets auto-close timers, and displays dynamic rotating QR codes.

Student Portal (Telemetry broadcaster)

Undergoes location gate validation, registers device fingerprint, verifies identity, and broadcasts active heartbeats.

Detailed Implemented Modules

1. UI Pill Icon Badges

Replaced text stats with graphical pills. Battery levels use green/yellow/red pill shapes showing percentage and charging states. Pulsing green/red location dots indicate live connection status.

2. Location GPS Gate

Applied geofence verification on student and teacher portals. The application blocks unauthorized usage with a full-screen red warning if location permissions are denied.

3. College Hours & Holiday Mode

Configured operating hours (e.g. 9:00 AM - 4:00 PM). Student and staff dashboards lock automatically outside these hours. Includes an HOD override for manual Holiday Mode locks.

4. Stable Browser Fingerprinting

Replaces random storage keys with stable browser hashes (combining timezone, screen, language, platform). Locks accounts to physical hardware and stops proxy cache resets.

5. Biometric WebAuthn Locking

Student WebAuthn biometrics are registered and cryptographically bound. Once registered, re-registration is locked to prevent security overrides unless explicitly reset by HOD.

6. Auto-Closing Sessions

Launches active sessions with a configurable timer. The server auto-closes the session once the timer runs out, preventing teachers from leaving classes open indefinitely.

7. Fake GPS Spoofing Detection

Tracks distance velocity, accuracy anomalies, and repeated coordinates to block mock location apps. Suspicious telemetry displays as LOCATION SUSPECT on dashboards.

8. GPS Dropout Warn & Escalation

Triggers alerts if a student disables location services during active periods. Coordinators can review the warning and escalate it to HOD dashboard feeds with one click.

9. Half-Day Attendance Logic

Accumulates checking minutes. Computes full-day ($\geq 75\%$ hours on campus), half-day (30-74% hours), or absent ($<30\%$ hours) metrics automatically.

Movement Timelines & System Security

3. Student Movement Timeline (Continuous Tracking)

The system records telemetry logs in a new database table (movement_events) to build a continuous path history. Each heartbeat ping records state shifts: CHECKIN, IN_CAMPUS, LEFT_CLASS, RETURNED, EXITED_CAMPUS, and GPS_DROPPED. The HOD can view a complete chronological timeline for any student directly inside the dashboard.

4. HOD Analytics Dashboard (Tab 5)

The HOD console is expanded with a fifth "Analytics" panel containing six pure-CSS graphical widgets. It compiles daily presence rings, top absentee lists, late arrival lists (check-in after 15 mins), frequent class leavers, class-wise comparative ratios, and 7-day presence trend lines.

5. OWASP Security Overhaul & Compliance

To secure institutional data, the system includes multiple security layers conforming to OWASP guidelines:

- JWT Access Tokens: Enforces secure token-based authentication on all API routes.
- Secure HTTPS: Runs HTTPS locally on startup with auto-generated self-signed certificates.
- Helmet WAF Headers: Protects against cross-site scripting (XSS) and injection exploits.
- Rate Limiting: Strict limits (15 requests/15 mins for logins) block brute-force and DDoS attempts.
- IDOR Protection: Validates user tokens against requested resources to prevent IDOR path manipulation.

